FDS EXPERIMENT-2

SUBIN P – 250701750

Upload and analyze the data set given in csv format and perform data preprocessing and visualization.

```
import pandas as pd

df = pd.read_csv('clean_sales_dataset.csv')
df['Date'] = pd.to_datetime(df['Date'])

pd.set_option('display.max_rows', None)
display(df)
```

	Serial no	Date	Product	Sales	Quantity	Region or place
0	1	2023-12-15	Monitor	261.18	1	West
1	2	2023-12-24	Laptop	5989.55	6	West
2	3	2023-07-20	Smartphone	9576.29	16	South
3	4	2023-08-14	Tablet	5248.49	13	East
4	5	2023-08-06	Headphones	1670.38	17	East
5	6	2023-01-17	Headphones	1708.08	17	West
6	7	2023-04-28	Headphones	1720.41	17	North
7	8	2023-09-22	Laptop	6003.06	6	North
8	9	2023-05-17	Headphones	195.65	2	North
9	10	2023-06-26	Monitor	1453.36	6	South
10	11	2023-03-11	Headphones	1436.12	15	South
11	12	2023-03-08	Smartphone	3058.45	5	North
12	13	2023-01-12	Monitor	2993.82	12	West
13	14	2023-01-12	Monitor	1741.73	7	North
14	15	2023-05-30	Headphones	1637.97	16	South
15	16	2023-08-17	Monitor	4509.30	18	South
16	17	2023-09-08	Headphones	705.39	7	South
17	18	2023-11-03	Headphones	314.03	3	East
18	19	2023-11-15	Monitor	4770.69	19	East
19	20	2023-11-24	Laptop	10010.62	10	North
20	21	2023-12-25	Monitor	1514.17	6	North
21	22	2023-07-10	Headphones	1389.38	14	East
22	23	2023-04-26	Headphones	1219.86	12	North
23	24	2023-12-25	Laptop	13972.62	14	South
24	25	2023-02-13	Headphones	1226.17	12	North
25	26	2023-02-02	Monitor	2762.75	11	West
26	27	2023-04-16	Laptop	2014.56	2	South
27	28	2023-05-13	Headphones	1348.99	13	South
28	29	2023-01-14	Laptop	5005.74	5	East
29	30	2023-10-27	Smartphone	9588.37	16	South
30	31	2023-03-26	Tablet	6008.84	15	North

31	32	2023-03-14	Monitor	4739.47	19	East
32	33	2023-08-27	Headphones	373.89	4	West
33	34	2023-10-11	Smartphone	7774.64	13	West
34	35	2023-01-24	Tablet	7211.27	18	South
35	36	2023-03-26	Tablet	1583.75	4	South
36	37	2023-05-14	Tablet	1574.01	4	South
37	38	2023-10-27	Smartphone	11410.81	19	South
38	39	2023-12-04	Monitor	4764.33	19	West
39	40	2023-09-10	Tablet	2377.68	6	East
40	41	2023-08-05	Monitor	249.40	1	South
41	42	2023-11-10	Monitor	1758.55	7	East
42	43	2023-06-27	Tablet	6821.06	17	West
43	44	2023-02-13	Tablet	7179.26	18	North
44	45	2023-09-15	Monitor	243.48	1	North
45	46	2023-08-23	Monitor	4565.47	18	South
46	47	2023-07-26	Headphones	1085.14	11	North
47	48	2023-09-17	Headphones	1085.51	11	East
48	49	2023-01-10	Smartphone	6576.87	11	North
49	50	2023-09-08	Smartphone	7803.01	13	East

```
import pandas as pd

df = pd.read_csv('clean_sales_dataset.csv')
display(df.head())
print(df.isnull().sum())
```

	Serial no	Date	Product	Sales	Quantity	Region or place
0	1	2023-12-15	Monitor	261.18	1	West
1	2	2023-12-24	Laptop	5989.55	6	West
2	3	2023-07-20	Smartphone	9576.29	16	South
3	4	2023-08-14	Tablet	5248.49	13	East
4	5	2023-08-06	Headphones	1670.38	17	East

```
Serial no          0
Date               0
Product            0
Sales              0
Quantity           0
Region or place    0
dtype: int64
```

```python
import pandas as pd

df = pd.read_csv('clean_sales_dataset.csv')
display(df.tail())
print(df.isnull().sum())
```

	Serial no	Date	Product	Sales	Quantity	Region or place
45	46	2023-08-23	Monitor	4565.47	18	South
46	47	2023-07-26	Headphones	1085.14	11	North
47	48	2023-09-17	Headphones	1085.51	11	East
48	49	2023-01-10	Smartphone	6576.87	11	North
49	50	2023-09-08	Smartphone	7803.01	13	East

```
Serial no          0
Date               0
Product            0
Sales              0
Quantity           0
Region or place    0
dtype: int64
```